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The M_1 model

OUTLINE

- Supersymmetric quantum mechanics
- The M_1 model
- Mapping to a solvable model (chain)

MAIN REFERENCES

[1] Edward Witten, *Constraints on supersymmetry breaking*, Nucl. Phys. B **202**, 235 (1982).

[2] Paul Fendley, Kareljan Schoutens, and Jan de Boer, *Lattice models with $\mathcal{N} = 2$ supersymmetry*, Phys. Rev. Lett. **90**, 120402 (2003).

[3] Paul Fendley, Bernard Nienhuis, and Kareljan Schoutens, *Lattice fermions with supersymmetry*, J. Phys. A: Math. Gen. **36**, 12399 (2003).

[4] Ricardo G. Dias, *Exact solution of the strong coupling $t - V$ model with twisted boundary conditions*, Phys. Rev. B **62**, 7791 (2000).

SUPERSYMMETRIC QUANTUM MECHANICS [1]

$\mathcal{N} = N$ supersymmetry:

- Supercharges Q_i $(i = 1, 2, \ldots, N)$
- Constraint 1: $Q_i^\dagger = Q_i$ (Hermitian / self-adjoint)
- Constraint 2: $Q_1^2 = Q_2^2 = \ldots Q_N^2$
- Constraint 3: $\{Q_i, Q_j\} = 0$ if $i \neq j$
- Hamiltonian: $H = Q_1^2 = Q_2^2 = \cdots = Q_N^2$

$\mathcal{N} = 2$ supersymmetry:

- $Q = (Q_1 + iQ_2)/\sqrt{2}$
- $Q^\dagger = (Q_1 - iQ_2)/\sqrt{2}$
- $H = \frac{1}{2}\{Q, Q^\dagger\}$

Equivalence of Hamiltonians:

$$2Q^\dagger Q = (Q_1 - iQ_2)(Q_1 + iQ_2) \tag{1}$$
$$= Q_1^2 + Q_2^2 + iQ_1Q_2 - iQ_2Q_1 \tag{2}$$

$$2QQ^\dagger = (Q_1 + iQ_2)(Q_1 - iQ_2) \tag{3}$$
$$= Q_1^2 + Q_2^2 - iQ_1Q_2 + iQ_2Q_1 \tag{4}$$

$\implies \{Q, Q^\dagger\} = \frac{1}{2}(Q_1^2 + Q_2^2) = H.$

Property 1: Q and $Q^\dagger$ are nilpotent:

$$\begin{aligned} Q^2 &= \frac{1}{2}(Q_1 + iQ_2)(Q_1 - iQ_2) && (5) \\ &= \frac{1}{2}(Q_1^2 - Q_2^2) && (6) \\ &= 0 && (7) \end{aligned}$$

Similar for $Q^\dagger$.

Property 2: all eigenvalues are non-negative

Note: not the approach of the reference, but probably more intuitive

$$\begin{aligned} H &= (Q^\dagger + Q)^2 && (8) \\ &= Q^\dagger Q + QQ^\dagger + (Q^\dagger)^2 + Q^2 && (9) \end{aligned}$$

Property 3: nonzero eigenstates come in degenerate doublets

Note: not the approach of the reference, but probably more intuitive (although less rigorous)

Let $\tilde{H} = Q^\dagger + Q$

Choice of basis: $Q^\dagger$ and Q are upper- and lower-diagonal, respectively.

Nilpotent: block form

$$Q^\dagger = \begin{pmatrix} 0 & M^\dagger \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 0 \\ M & 0 \end{pmatrix} \tag{10}$$

Matrix M is of size $n \times m$ (does not have to be square). Intuition: particle-hole symmetry.

Eigenvalue equation can be written as

$$\begin{pmatrix} 0 & M^\dagger \\ M & 0 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = E \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \tag{11}$$

Each eigenvalue $E > 0$ comes with an eigenvalue $-E$

$$\begin{pmatrix} 0 & M^\dagger \\ M & 0 \end{pmatrix} \begin{pmatrix} \psi_1 \\ -\psi_2 \end{pmatrix} = -E \begin{pmatrix} \psi_1 \\ -\psi_2 \end{pmatrix} \tag{12}$$

Orthogonality: $|\psi_1|^2 = |\psi_2|^2 = 1/2$.

The spectrum is thus symmetric around energy zero.

Squaring the Hamiltonian:

$$\begin{pmatrix} 0 & M^\dagger \\ M & 0 \end{pmatrix} \begin{pmatrix} 0 & M^\dagger \\ M & 0 \end{pmatrix} = \begin{pmatrix} M^\dagger M & 0 \\ 0 & MM^\dagger \end{pmatrix} \tag{13}$$

Degenerate eigenstates $|\psi_1\rangle$ and $|\psi_2\rangle$ with energy E^2

Upper-left (lower-right) block: dimension $m \times m$ ($n \times n$)

If $m > n$ ($n > m$): there are $m - n$ ($n - m$) zero-energy states $|\psi_1\rangle$ ($|\psi_2\rangle$)

THE $\mathbf{M_1}$ MODEL [2,3]

M_1 models: spinless fermions hopping on a lattice / graph, subject to a nearest-neighbor blockade:

$$H_{M_1} = Q^\dagger Q + QQ^\dagger, \tag{14}$$

with

$$Q^\dagger = \sum_i P_{\langle i \rangle} c_i^\dagger, \qquad Q = \sum_i P_{\langle i \rangle} c_i \tag{15}$$

The projectors $P_{\langle i \rangle}$ forbid two neighboring sites to be occupied simultaneously:

$$P_{\langle i \rangle} = \prod_{j \in \langle i \rangle} \left(1 - c_j^\dagger c_j\right) \tag{16}$$

Some basic properties:

• Explicit Hamiltonian:

$$H_{M_1} = \sum_i \left[\sum_{j \in \langle i \rangle} P_{\langle i \rangle} c_i^\dagger c_j P_{\langle j \rangle} \right] + \sum_i P_{\langle i \rangle} \tag{17}$$

•Square of a PXP-like fermion model:

$$H_{\mathrm{PXP}} = Q^\dagger + Q \tag{18}$$

$$= \sum_i P_{\langle i \rangle} (c_i^\dagger + c_i) \tag{19}$$

Some interesting results on this upcoming

•Fermion number conservation:

$$[H_{M_1}, F] = 0, \qquad F = \sum_i c_i^\dagger c_i \tag{20}$$

• $\mathcal{N} = 2$ supersymmetry!

MAPPING TO A SOLVABLE MODEL (CHAIN)

Focus in what follows: one-dimensional chain.

Consider (17), but ignore the second term for the moment. Call it H_0. This is a $t-V$ chain in the strong-coupling limit. Solved in [4] by mapping to a tight-binding chain.

Mapping to tight-binding chain:

1. $\circ\bullet$ maps to an occupied site $\bullet$
2. remaining empty sites map to empty sites

An N-site chain maps to chain of length $L = N - f$ (f is the number of fermions):

$$H_0 \to \sum_{i=1}^{L-1} \left[d_i^\dagger d_{i+1} + T^{-1} d_1^\dagger d_L \right] + \text{H.c.} \tag{21}$$

Here: $d_i^\dagger$ (d_i) create (annihilate) a fermion on site i. Second term is to account for hoppings between last and first site.

Single fermion

Note: in H_{M_1}, all projections evaluate to unity $\implies$ trivial, the more difficult approach followed here can be used as well in the case of multiple fermions.

Over-complete set of states:

$$|i,k\rangle = \frac{1}{\sqrt{L}} \sum_{j=1}^{L} e^{ikj} T^{j-1} |i\rangle \tag{22}$$

Then:

$$\sum_{i,k} |i,k\rangle H \langle i,k| = \sum_k \left[\sum_{j=1}^{L-1} \left(|i+1,k\rangle\langle i,k| + e^{ik} |1,k\rangle\langle L,k| \right) + \text{H. c.} \right] \tag{23}$$

This is a tight-binding chai with a flux e^{ik}.

Eigenvalues:

$$E(k,\tilde{k}) = 2\cos\left(\tilde{k} - \frac{k}{L}\right) \tag{24}$$

Not all eigenstates correspond to eigenstates of H_0 because the set of states is over-complete. Requirement: eigenstates need to have the proper momentum:

$$\tilde{k} - \frac{k}{L} = k \pmod{2\pi} \tag{25}$$

Multiple fermions

Non-interacting model, eigenstates are sets of single-particle states with momenta $\tilde{k}_i - k/L$ ($L = N - f$).

Requirement:

$$\sum_{i=1}^{f} \left(\tilde{k}_i - \frac{k}{L} \right) = k \pmod{2\pi} \tag{26}$$

Adding the interaction term

$$V = \sum_i P_{\langle i \rangle} \tag{27}$$

$$= \sum_i \left(1 - c_{i-1}^\dagger c_{i-1}\right)\left(1 - c_{i+1}^\dagger c_{i+1}\right) \tag{28}$$

$$= L - 2f + \sum_i c_{i-1}^\dagger c_{i-1} c_{i+1}^\dagger c_{i+1} \tag{29}$$

We ignore the constant $L - 2f$ in what follows.

Mapping:

$$V \to \sum_i d_i^\dagger d_i d_{i+1}^\dagger d_{i+1} \tag{30}$$

Note: flux (boundary twist does not affect).

Jordan-Wigner transformation (mapping to an integrable model) [2,3]

Jordan-Wigner transformation:

$$H \to \frac{1}{2} \sum_j \left[\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y - \Delta \sigma_j^z \sigma_{j+1}^z \right] \tag{31}$$

with $\Delta = -1/2$. Twisted boundary conditions ($t = e^{ik}$):

$$\sigma_{L+1}^+ = (-1)^{f+1} t \sigma_1^+ \tag{32}$$

$$\sigma_{L+1}^- = (-1)^{f+1} t^{-1} \sigma_1^- \tag{33}$$

Bethe-Ansatz solution [2,3]

f coupled equations:

$$t^{-1} (\mu_k)^{N-f} = \prod_{j=1}^{f} \frac{\mu_k \mu_j + 1 - \mu_k}{\mu_k \mu_j + 1 - \mu_j} \tag{34}$$

Eigenvalues:

$$E = N - 2f + \sum_{k=1}^{f} \left[\mu_k + \frac{1}{\mu_k} \right] \tag{35}$$

$$= N - 2f + 2 \sum_{k=1}^{f} \cos(\phi_k) \tag{36}$$

for $\mu_k = e^{i\phi_k}$. Sum of momenta must be the right value (encoded in t)

Comments:

- If $\mu_1, \mu_2, \ldots, \mu_f$ is a solution with f fermions, then $\mu_1, \mu_2, \ldots, \mu_f, 1$ is a solution with $f+1$ fermions having the same energy $\implies$ degeneracies
- Degenerate μ_k's not allowed ("eigenstate vanishes when interchanging two particles"). Exception: $\mu_j = \mu_k = \exp(\pm i\pi/3)$ as then ratio evaluates to $0/0$.

Eigenstates:

$$|\psi\rangle = B(\mu_1) \cdots B(\mu_k)|0\rangle. \tag{37}$$

Matrix product state. Interesting: low bond dimension for $\exp(\pm i\pi/3)$